# R118 | Realized TRAIN Computation Assets (Not A Selector Result)

18/18 problems; 18/89 deployments. Frozen input-only TRAIN selection, seed2. No holdout answers or label changes.

| Task | Native method | Cases | Nodes | Steps | Starts | Active prob rows | Cost close |
|---|---|---|---|---|---|---|---|
| TSP | LEHD | 8 | 52-496 | 52-496 | 1-1 | 2144 | 8/8 |
| CVRP | RELD_CVRP | 8 | 102-472 | 114-494 | 100-100 | 220303 | 8/8 |
| ATSP | MATNET | 8 | 20-92 | 20-92 | 20-92 | 30214 | 0/8 |
| OVRP | reld_mt | 8 | 50-98 | 61-113 | 50-98 | 45025 | 8/8 |
| VRPB | reld_mt | 8 | 51-97 | 58-106 | 40-77 | 39444 | 8/8 |
| VRPL | reld_mt | 8 | 57-99 | 66-111 | 57-99 | 59648 | 8/8 |
| VRPTW | reld_mt | 8 | 54-100 | 67-120 | 54-100 | 62160 | 8/8 |
| OVRPTW | reld_mt | 8 | 54-98 | 67-117 | 54-98 | 54536 | 8/8 |
| OVRPB | reld_mt | 8 | 54-96 | 63-107 | 43-76 | 41946 | 8/8 |
| OVRPL | reld_mt | 8 | 51-97 | 60-109 | 51-97 | 51898 | 8/8 |
| VRPBL | reld_mt | 8 | 50-97 | 57-106 | 40-77 | 35281 | 8/8 |
| VRPBTW | reld_mt | 8 | 55-99 | 71-124 | 44-79 | 51072 | 8/8 |
| VRPLTW | reld_mt | 8 | 57-100 | 68-119 | 57-100 | 59685 | 8/8 |
| OVRPBL | reld_mt | 8 | 56-98 | 64-109 | 44-78 | 38573 | 8/8 |
| OVRPBTW | reld_mt | 8 | 55-99 | 71-118 | 44-79 | 44715 | 8/8 |
| OVRPLTW | reld_mt | 8 | 50-96 | 67-112 | 50-96 | 49266 | 8/8 |
| VRPBLTW | reld_mt | 8 | 55-97 | 70-118 | 44-77 | 44235 | 8/8 |
| OVRPBLTW | reld_mt | 8 | 52-99 | 70-117 | 41-79 | 43598 | 8/8 |

Actual native recipe; fresh per-source RNG, not historical process replay. Forced distributions excluded; masks are not independent certificates.

# What This Preparation Does And Does Not Establish

Captured: every actual action, node/trajectory identity, real decoder output and pre/post state, not trajectory means.

All native starts retained. TSP/ATSP execute native augmentation budgets but serialize original orientation0 used by the method.

Probabilities and alternative-start histories are observed teacher computation, not prefixes fabricated from the best final tour.

States recoverable from input and full routes are not new independent information. Steps are not independent instances.

One initial CVRP observer call failed on missing POMO_IDX. It is preserved; only uncompleted B was explicitly recovered.

Recovery derives column-index metadata without writing native state; physical start nodes remain actual recorded actions.

The CPU observer bound rose384MiB->1GiB before recovery; inputs, weights, solver budgets and original labels did not change.

Native cost close to frozen TRAIN: 136/144; fully matching task samples: 17/18. MATNET: 0/8.

Native MATNET role inputs were recorded, but the cause of its cost difference is not isolated; no label-error claim.

No selector trained, checkpoint selected, TEST/LIB evaluated or zero-selection improvement claimed by this asset release.

No cheap sufficient state, learnability, Markov compression or decision-transfer benefit has been established.

R58 semantics remain deferred. Current legal pools and single reld_mt remain unchanged.

The long-running whole18 improvement goal remains active. New NN training is not automatically approved by capture completion.